Getting the Joke: Stacking Diverse Transformers for Humor Intensity Regression in Edited News Headlines

Jannatul Naima[1], Trina Chakraborty[2,⋆,†], Swapnil Chowdhury[3,⋆], Mst Tuly Khatun[3,⋆⋆], Monalisa Tripura[3,⋆⋆], and Farjana Alam Tofa[3,⋆⋆]

[1] University of Chittagong, Chittagong
[2] Shahjalal University of Science and Technology, Sylhet
[3] Chittagong University of Engineering and Technology, Chittagong
jannatulnaima.cse.cu@gmail.com, trinasust5002@gmail.com, swapnilchy26@gmail.com, u1904035@student.cuet.ac.bd, monalisatrip24@gmail.com, farjanaalamtofa12@gmail.com

Abstract. Humor is difficult to model computationally because it is subjective, context dependent, and often produced by a very small change in wording. We present a two-stage stacking ensemble for predicting the funniness of micro-edited news headlines in SemEval-2020 Task 7 (Subtask 1). The proposed framework combines four architecturally diverse transformer encoders, namely DeBERTa-v3, RoBERTa, StructBERT, and Sentence-BERT (MPNet-base), under a marked dual-input representation that explicitly identifies the edited token and preserves the contrastive relationship between the original and modified headlines. Base-model predictions are integrated through a leakage-free meta-regressor trained using 5-fold out-of-fold cross-validation, so that no meta-training instance is produced by a model that observed its label. An exhaustive search over all 70 four-model combinations is used to assess how architectural diversity affects ensemble behaviour, and three fusion strategies are compared. The final system achieves a test root mean square error (RMSE) of 0.4858 on the Humicroedit benchmark, which is lower than the alternative fusion strategies evaluated here and than the previously published SemEval-2020 systems included in our comparison. All results are from single training runs, so small differences between the top configurations should be read as comparable rather than strictly ordered.

Keywords: Computational Humor · Transformer-Based Regression · Stacking Ensemble · Edited News Headlines

1 Introduction

Humor is a fundamental component of human communication that relies on linguistic creativity, contextual reasoning, and shared world knowledge. Despite

† Corresponding author.
⋆ The second and third authors contributed equally to this work.
⋆⋆ The fourth, fifth, and sixth authors contributed equally to this work.

substantial progress in natural language processing, computational humor assessment remains difficult because humor is highly subjective, context-dependent, and culturally variable. To support research in fine-grained humor prediction, Hossain et al. [10] introduced the Humicroedit dataset, consisting of approximately 15,000 English news headlines from Reddit modified through single-word edits and annotated with continuous funniness scores ranging from 0 to 3. Building on this resource, SemEval-2020 Task 7 [11] defined two subtasks: funniness regression and pairwise comparison of edited headlines.

Fine-grained humor prediction has value beyond the shared-task setting. Systems that rank or filter user-generated content, assist headline writing, or support conversational agents need an estimate of how funny a piece of text is likely to be, not merely a binary judgement of whether it was intended as a joke. The Humicroedit setting isolates this problem in an unusually controlled way, because the original and edited headlines differ by exactly one content word. Any change in the annotated funniness score can therefore be attributed to that substitution and to its interaction with the surrounding context. This makes the dataset a useful testbed for studying how well pretrained encoders capture semantic incongruity, and at the same time makes the task hard: the signal separating a funny edit from a flat one is carried by a single token whose effect depends on the whole headline.

While transformer-based models have shown strong performance on humor-related NLP tasks, modeling the effect of a single-word edit on perceived funniness remains challenging. In Humicroedit, a single edited word can change the meaning and humor of a headline in subtle, context-dependent ways, making it important to identify the location of the edit for fine-grained regression. Ensemble methods can also benefit from combining complementary transformer representations, but the effects of architectural diversity, input representation, and fusion strategy have not been examined together in this setting. Motivated by these issues, we investigate a compact transformer ensemble that explicitly identifies the edited token and combines complementary model representations through leakage-free stacking.

Two design questions follow from this framing. The first concerns how the edit should be presented to the encoder. Supplying only the edited headline requires the model to recover the contrast without knowing what was replaced, whereas supplying both headlines without marking the substitution leaves the model to locate the change on its own. The second concerns how several encoders should be combined. Earlier systems on this benchmark obtained strong results by aggregating large numbers of fine-tuned models, so it remains open whether comparable accuracy can be reached with a small set of encoders chosen for architectural diversity, and whether a learned second-stage regressor is preferable to fixed averaging.

We address Subtask 1 using a stacking ensemble that combines four complementary transformer encoders, namely RoBERTa-large [15], DeBERTa-v3-large [9], StructBERT-large [31], and Sentence-BERT (MPNet-base; hereafter Sentence-BERT) [22], under a unified marked dual-input representation. Unlike

prior large-scale ensembles that rely primarily on model quantity, our approach focuses on carefully selected architectural diversity combined with leakage-free meta-learning. Through experiments on the Humicroedit benchmark, we evaluate the effects of input representation, model composition, and fusion strategy. In particular, our final ensemble achieves a root mean square error (RMSE) of 0.4858, which is lower than the best previously reported system included in our comparison (Hitachi ensemble, RMSE 0.4959).

The key contributions of this work are as follows:

1. We investigate a structured input encoding strategy for edited headlines, introducing explicit edit markers that help transformer models identify the substitution location and model the semantic contrast relevant to humor regression.
2. We propose a compact transformer stacking ensemble combining DeBERTa-v3-large, RoBERTa-large, StructBERT-large, and Sentence-BERT (MPNet-base), and investigate how architectural diversity contributes to ensemble performance.
3. We develop a leakage-free stacking framework with 5-fold out-of-fold meta-learning and conduct an exhaustive evaluation of fusion strategies and four-model combinations, achieving a test RMSE of 0.4858 on the Humicroedit benchmark.

2 Related Work

Humor understanding in text has evolved from binary classification and ranking based approaches toward fine-grained regression-based modeling. Earlier studies primarily focused on detecting humor or comparing relative funniness, while recent studies predict continuous humor intensity scores. This shift comes with the Humicroedit dataset [10], containing minimally edited news headlines with continuous funniness scores. Building on it, Hossain et al. [11] defined humor assessment as a regression task in SemEval-2020 Task 7. Participating systems for Subtask 1 included feature-based models (lexical, semantic, stylistic features), traditional regression, neural networks, transformer-based models like BERT and RoBERTa, and ensemble approaches.

Early computational humor relied on hand-crafted linguistic and statistical features such as n-grams and stylistic patterns. The #HashtagWars shared task [21] treated humor as a ranking problem, acknowledging its graded nature. Systems used feature-based methods with traditional machine learning, while Yan and Pedersen [33] demonstrated the effectiveness of n-gram language models. Beyond shared tasks, Reyes et al. [23] explored linguistic humor detection, and Barbieri and Saggion [1] investigated irony and humor on Twitter. Several SemEval-2020 Task 7 systems (e.g., LT3 [29], Buhscitu [13], ECNU [35], Hasyarasa [6]) used feature-based approaches, achieving mid-to-lower rankings. Though interpretable, these methods require manual effort, are domain-specific, and struggle with deep contextual nuances in edited text.

Deep learning advanced humor detection and ranking. Chen and Soo [4] proposed a convolutional neural network (CNN) with pretrained embeddings, while Bertero and Fung [2] used neural representations for humor ranking. Miraj and Aono [19] combined a bidirectional long short-term memory (LSTM) network, or BiLSTM, with pretrained sentence embeddings. Dick et al. [8] used ridge regression with rich features and an auxiliary LSTM network with GloVe (Global Vectors for Word Representation) embeddings. Complementary resource efforts, including FunLines [12] for humorous headline collection and the HAHA corpus [5] for Spanish, broadened the data available for humor modelling beyond English, as did the BERT-based scoring approach of Mao and Liu [18]. Among SemEval-2020 systems, YNU-HPCC [28] used a bidirectional gated recurrent unit (Bi-GRU) ensemble, but its performance lagged behind transformer-based ensembles.

With transformers, humor understanding improved via better contextual representations. Pretrained models like BERT [7] excel without handcrafted features. For edited headlines, Jin et al. [14] used a contrastive incongruity approach, while Mahurkar and Patil [17] evaluated multiple transformer variants, showing strong generalization. Ensemble transformers performed well: the Hitachi system [20] used sentence-pair regression with a large ensemble of BERT, GPT-2, RoBERTa, XLNet, Transformer-XL, and XLM, with RoBERTa as the best individual model. The Amobee system [24] combined 90 fine-tuned BERT, XLNet, and RoBERTa models, achieving second place. Other works include BFHumor [25], which combines a BERT regressor with the Flair library, and Soloveva [26], who fed BERT embeddings into support vector regression (SVR) and logistic regression.

Transformer methods outperform earlier feature-based and conventional neural approaches for fine-grained humor prediction. However, many systems rely on single-model architectures, missing complementary information. Deep learning models often overlook fine-grained stylistic and incongruity cues. Ensemble learning, especially stacking, remains underexplored for regression-based humor prediction. To address this, we propose a stacking ensemble for SemEval-2020 Task 7 Subtask 1, combining predictions from multiple base models into a meta-regressor.

3 Proposed Framework

We propose a two-stage stacking ensemble for continuous humor intensity regression on the Humicroedit dataset. The framework is designed around the hypothesis that humor perception depends on heterogeneous linguistic signals, including lexical substitution effects, semantic plausibility, and contextual incongruity. To capture these complementary factors, we combine multiple transformer encoders with a leakage-free stacking mechanism that aggregates their predictions at a second stage. The overall architecture is illustrated in Figure 1.

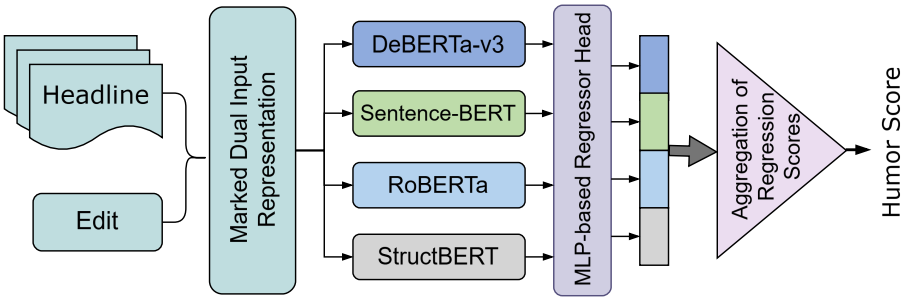


Fig. 1. Overview of the proposed framework. Model names are abbreviated in the figure; the encoders are DeBERTa-v3-large, StructBERT-large, Sentence-BERT (MPNet-base), and RoBERTa-large.

Input Representation

Each instance in the Humicroedit dataset consists of a news headline in which a single content word is substituted to introduce humor. Since humor in this setting arises from the semantic contrast between the original and edited word, explicitly preserving the edit location is important. We construct a *marked dual-input* representation to explicitly encode this original-edited contrast rather than leaving the model to infer it implicitly.

In this representation, the substituted token in the original headline is wrapped with special markers `<edit>` and `</edit>` to explicitly indicate the modification site and is paired with the corresponding edited headline using a `[SEP]` separator. This yields a structured input of the form:

> *Headline:* Kushner to visit Mexico following latest Trump tirades.
>
> *Edit:* therapist
>
> *Input representation:* Kushner to visit `<edit>Mexico</edit>` following latest Trump tirades `[SEP]` Kushner to visit therapist following latest Trump tirades

This design preserves both original and modified contexts while explicitly exposing the substitution position, enabling cross-segment self-attention to model the semantic shift induced by the edit. The special tokens are added to the tokenizer vocabulary, and all sequences are truncated or padded to a maximum length of 128 tokens. We also normalized inconsistent markup for edited tokens (e.g., `<word/>` or `<word>`).

Fine-Tuning Transformer Models

We fine-tune four transformer encoders: DeBERTa-v3-large[4] [9], StructBERT-large[5] [31], Sentence-BERT (MPNet-base)[6] [22], and RoBERTa-large[7] [15]. These models are chosen for their complementary characteristics spanning disentangled attention, structural encoding, sentence-level representation learning, and contextual pretraining. This architectural diversity is intended to provide complementary predictions, since ensemble methods can benefit when constituent models make different errors [3].

Each model receives the tokenized marked dual-input sequence and passes it through the transformer encoder. For all four models, the hidden state of the `[CLS]` token from the final encoder layer serves as the aggregate sequence representation. We adopt CLS pooling uniformly across models, including the Sentence-BERT checkpoint, rather than its conventional sentence-embedding pooling strategy. This choice provides a consistent aggregation mechanism across the four encoders when processing the marked dual-input representation. The resulting vector is fed into a two-layer multi-layer perceptron (MLP) regression head:

$$\hat{y} = \mathbf{W}_2 \cdot \mathrm{GELU}(\mathrm{Dropout}_{0.1}(\mathbf{W}_1 \cdot \mathrm{Dropout}_{0.3}(\mathbf{h}_{\mathrm{CLS}}))) \tag{1}$$

where $\mathbf{W}_1 \in \mathbb{R}^{256\times d}$ projects the encoder output to 256 dimensions and $\mathbf{W}_2 \in \mathbb{R}^{1\times 256}$ produces the final scalar score. The encoder and regression head are trained jointly end-to-end, allowing the encoder representations to adapt specifically to the humor regression objective. Final predictions are clipped to $[0, 3]$ at inference to conform to the annotated label range.

Fusion of Transformer Model Scores

Integrating predictions from multiple independently trained transformer models is an effective strategy for improving robustness and overall regression performance compared to relying on a single model. To leverage the diversity of the fine-tuned models' learned representations, we aggregate their predicted funniness regression scores. We specifically investigate three ensemble fusion strategies: mean blending, weighted averaging, and stacking.

Mean blending assigns uniform importance to all base-model predictions, while weighted averaging assigns higher weights to models with lower validation RMSE. For stacking, we train a second-stage meta-regressor on the prediction vectors produced by the base models. To avoid label leakage, we adopt a strict 5-fold out-of-fold (OOF) protocol. In each fold, the base models are trained on four folds and generate predictions for the held-out fold only. The OOF predictions from all folds are concatenated to form the meta-training set, ensuring that

[4] `sileod/deberta-v3-large-tasksource-nli`
[5] `bayartsogt/structbert-large`
[6] `sentence-transformers/all-mpnet-base-v2`
[7] `FacebookAI/roberta-large`

each training instance for the meta-learner is produced by models that did not observe its ground-truth label during fine-tuning. At inference time, each base model generates predictions for the test set independently. These predictions are then used as inputs to the trained meta-regressor to produce final outputs. To ensure robustness, we evaluate multiple meta-regression algorithms, including Linear Regression, Bayesian Ridge, Automatic Relevance Determination (ARD), SVR with a Radial Basis Function (RBF) kernel, TheilSen Regression, Decision Tree, and Passive Aggressive Regression.

4 Experiments and Evaluation

4.1 Dataset, Evaluation Metrics, and Experimental Setup

We evaluate our approach on the Humicroedit benchmark [11] (SemEval-2020 Task-7, Subtask-1), which consists of 9,652 training, 2,419 development, and 3,024 test instances. Each instance contains a news headline, a single-word edit applied to that headline, and a corresponding mean funniness score (*meanGrade*) in the range 0–3, obtained by averaging multiple human annotations. The edited headlines are constructed by replacing one content word with a semantically or phonetically related substitute designed to introduce humorous incongruity.

The primary evaluation metric is RMSE over the full test set (lower is better). Following the original task formulation, we also report RMSE on the union of the top-N% and bottom-N% headlines by funniness score ($N \in \{10, 20, 30, 40\}$), which evaluates discrimination at humor extremes.

All models are optimized with AdamW [16] using a learning rate of 2×10^{-5}, weight decay of 0.01, and gradient clipping at norm 1.0, with a cosine annealing learning rate schedule. DeBERTa-v3-large is trained at a reduced learning rate of 1×10^{-5} due to its larger parameter count and sensitivity to optimization dynamics. Training runs for a maximum of five epochs with early stopping based on validation RMSE with a patience of two epochs, ensuring that the best checkpoint is retained. The mean squared error (MSE) loss is used as the training objective across all models.

4.2 Results and Analysis

We structure our evaluation around four research questions that directly motivated the design choices in our framework:

- **RQ1:** How does the choice of input representation affect humor regression, and why is the marked dual-input representation adopted?
- **RQ2:** Which combination of base transformer models yields the strongest stacking performance, and how does the proposed four-model set compare with alternative four-model compositions?
- **RQ3:** Does learned stacking outperform simpler combination strategies such as mean blending and weighted averaging?
- **RQ4:** How does the proposed model compare with previously published systems on the Humicroedit benchmark in terms of regression performance?

Input Representation and Design Rationale (RQ1) Before selecting models or fusion strategies, we consider the input representation used for humor regression. This consideration is particularly relevant to Humicroedit, where humor is introduced through the substitution of a single word and therefore depends on the contrast between the original and edited forms. Providing only the edited headline removes the original wording and requires the model to infer the transformation without an explicit indication of what changed.

We considered four candidate input formats during model development: *edited* (edited headline only), *concat* (original headline paired with the edit word via `[SEP]`), *both* (original and edited headlines separated by `[SEP]` without explicit change marking), and *marked* (original headline with the substituted token enclosed in `<edit>` and `</edit>` tags, followed by the edited headline).

The *edited* format removes the original wording and therefore does not explicitly preserve the contrast introduced by the edit. The *concat* and *both* formats preserve more of this information but do not explicitly identify the location of the substitution. In contrast, the *marked* format provides an explicit structural cue by localizing the substituted token with `<edit>` and `</edit>` markers while retaining both the original and edited headlines. We therefore use the *marked* representation in the proposed framework because it directly exposes the edit location and preserves the original-edited contrast for subsequent transformer modeling.

We note a limitation of this analysis. The four formats were compared informally during model development, but not under the controlled training configuration used for the results reported in Section 4.1, and we therefore do not report those preliminary numbers here. Re-running all four formats under identical settings requires fine-tuning large encoders several times over, which was beyond the computational budget available for this study. The marked representation is adopted on representational grounds rather than on the basis of a controlled ablation, and a like-for-like comparison of the four formats under identical settings is left to future work.

Base Model Selection and Combination Analysis (RQ2) With the input representation fixed, we next select transformer base learners. We fine-tune eight candidates under identical settings—the four models introduced above (DeBERTa-v3-large, StructBERT-large, Sentence-BERT, RoBERTa-large) together with BERT, XLNet, PhraseBERT, and MPNet (standalone)—with individual results reported in Table 1. Note that MPNet (standalone) is a directly fine-tuned MPNet checkpoint and is distinct from Sentence-BERT, which is a Sentence-BERT model built on an MPNet-base checkpoint.

A naive top-k selection by test RMSE would choose DeBERTa-v3, BERT [7], XLNet [34], and Sentence-BERT [22]. However, secondary metrics reveal clear weaknesses in this naive selection: XLNet is worst at $N10\%$ (0.8708), indicating poor behavior on extreme cases, and although BERT [7] achieves a lower overall test RMSE than StructBERT [31] and Sentence-BERT [22], it underperforms both at the funniness extremes ($N10\%$: 0.7962 vs. 0.7721 and 0.7864; $N20\%$:

Table 1. Individual model performance (RMSE; lower is better). Secondary metrics report RMSE on the top+bottom $N\%$ by funniness score.

Model	Dev	Test	N10%	N20%	N30%	N40%
DeBERTa-v3* [9]	**0.5098**	**0.5052**	**0.7267**	**0.6276**	**0.5694**	**0.5322**
BERT [7]	0.5293	0.5290	0.7962	0.6905	0.6192	0.5674
XLNet [34]	0.5288	0.5365	0.8708	0.7381	0.6501	0.5856
Sentence-BERT* [22]	0.5478	0.5444	0.7864	0.6840	0.6196	0.5769
PhraseBERT [30]	0.5417	0.5459	0.8268	0.7140	0.6373	0.5866
StructBERT* [31]	0.5569	0.5463	0.7721	0.6793	0.6158	0.5755
MPNet (standalone) [27]	0.5571	0.5565	0.8626	0.7414	0.6583	0.6007
RoBERTa* [15]	0.5707	0.5644	0.7982	0.7028	0.6380	0.5959

* Selected for stacking.
Sentence-BERT is the Sentence-BERT model on the MPNet-base checkpoint (`sentence-transformers/all-mpnet-base-v2`); MPNet (standalone) is a separate, directly fine-tuned MPNet checkpoint and is not the same model.

0.6905 vs. 0.6793 and 0.6840), where correctly identifying strongly funny or unfunny headlines matters most.

We therefore consider both individual performance and architectural diversity when forming the candidate ensemble. The final set comprises DeBERTa-v3-large (disentangled attention), StructBERT-large (structural encoding), Sentence-BERT (sentence-level representation learning), and RoBERTa-large [15] (contextual pretraining). These models encode different inductive biases, yielding complementary error patterns that are beneficial for stacking ensembles [3]. This four-model set is motivated by architectural diversity, while the exhaustive combination search provides an empirical assessment of its performance relative to alternative four-model compositions.

Table 2. Representative four-model combinations from the exhaustive $\binom{8}{4} = 70$ search (Linear Regression stacking with 5-fold OOF). Sentence-BERT is the Sentence-BERT model on the MPNet-base checkpoint; MPNet is the separate, standalone MPNet checkpoint.

Model Combination	Dev RMSE	Test RMSE
DeBERTa + StructBERT + Sentence-BERT + RoBERTa	**0.4907**	**0.4858**
DeBERTa + StructBERT + Sentence-BERT + MPNet	0.4906	0.4862
DeBERTa + StructBERT + Sentence-BERT + BERT	0.4908	0.4862
DeBERTa + StructBERT + RoBERTa + BERT	0.4906	0.4862
DeBERTa + StructBERT + Sentence-BERT + XLNet	0.4899	0.4878
StructBERT + Sentence-BERT + RoBERTa + BERT	0.5067	0.5020
MPNet + XLNet + PhraseBERT + Sentence-BERT	0.5108	0.5149

We validate this choice via exhaustive search over all $\binom{8}{4} = 70$ combinations using linear regression stacking with 5-fold out-of-fold validation [32]. Results (Table 2) show that the proposed set achieves the best observed test RMSE of 0.4858, while several close variants differ by only 0.0004. DeBERTa-v3-large and StructBERT-large appear consistently among the strongest configurations, suggesting that their complementary representations contribute positively to ensemble performance. As all reported results are from single training runs, differences on the order of 0.0004–0.001 are within the range plausibly attributable to run-to-run variance (e.g., weight initialization, batch ordering), and should therefore be interpreted as indicating comparable performance among top configurations rather than a strict ranking.

Removing DeBERTa-v3-large leads to a substantial degradation: the best non-DeBERTa combination achieves 0.5020, a difference of 0.0162 from the proposed model, while the weakest evaluated configuration reaches 0.5149. XLNet variants also underperform when included with DeBERTa-v3-large, consistent with their weaker performance on extreme-score cases. Overall, the results suggest that ensemble performance depends not only on individual model strength but also on the complementary characteristics of the constituent models.

Fusion Strategy Analysis and Comparison with Prior Systems (RQ3, RQ4) Having established the input representation and base model pool, we evaluate fusion strategies and compare our proposed method against prior SemEval-2020 Task 7 systems. Table 3 summarizes all results.

Table 3. Comparison with published SemEval-2020 Task 7 systems and evaluated fusion strategies (RMSE; lower is better). Stacking uses 5-fold OOF cross-validation.

System	Dev RMSE	Test RMSE
HumorAAC [8]	—	0.5285
BFHumor [25]	—	0.5199
Amobee [24]	—	0.5029
Hitachi [20]	—	0.4959
Best individual model (DeBERTa)	0.5098	0.5052
Mean Blending	0.5129	0.5060
Weighted Averaging	0.5118	0.5050
Stacking – TheilSen	0.4919	0.4872
Stacking – SVR (RBF)	0.4995	0.4903
Stacking – ARD Regression	0.4908	0.4858
Stacking – Bayesian Ridge	0.4907	0.4858
Stacking – Linear Regression (proposed)	**0.4907**	**0.4858**

We first evaluate simple fusion strategies. Mean blending achieves 0.5060 test RMSE, while weighted averaging slightly improves to 0.5050, but both are limited by fixed global weights that cannot adapt to instance-level model reliability.

Learned stacking improves performance over the simpler fusion strategies evaluated in this study. Linear Regression, Bayesian Ridge, and ARD Regression all achieve the same observed test RMSE of 0.4858, indicating that the final performance is not sensitive to the choice among these three meta-regressors in this experiment. We note that these results reflect single runs rather than multiple seeds, so we treat the equivalence among these three meta-regressors as indicative rather than statistically confirmed. SVR (0.4903) and TheilSen (0.4872) underperform. We adopt linear stacking with 5-fold OOF validation as the final fusion method because it provides the same observed performance with a simple and interpretable formulation.

Compared to prior systems, the best published result (Hitachi [20], 0.4959) is surpassed by all evaluated stacking variants, with the proposed model achieving 0.4858 test RMSE. These results demonstrate that the proposed stacking framework achieves a lower observed RMSE than the prior systems included in our comparison and the evaluated individual models.

5 Conclusion

This work investigates the use of architectural diversity in a compact transformer stacking ensemble for humor intensity regression on SemEval-2020 Task 7. By combining four complementary transformer encoders within a leakage-free stacking framework, the proposed system achieves a test RMSE of 0.4858, outperforming the previously published systems included in our comparison.

Our experiments further indicate that explicitly marking the edited token provides a useful representation for modeling the contrast between original and modified headlines. The combination analysis also shows that ensemble performance depends on both individual model quality and the complementary characteristics of the constituent models, while out-of-fold training provides a leakage-free procedure for learning the meta-regressor.

The primary remaining challenge lies in modeling highly humorous headlines, where severe class imbalance and stylistic variability limit reliable estimation. Future work should investigate targeted data augmentation, humor-aware contrastive objectives, and distribution-sensitive optimization strategies to better capture rare but strongly humorous edits. A further limitation is that all reported results come from single training runs, since repeatedly fine-tuning eight large transformer encoders was beyond the computational resources available to us. Replication across multiple random seeds, with standard deviations and significance testing, together with a controlled comparison of the four candidate input formats, would allow the small margins between the strongest configurations to be interpreted with more confidence.

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